

Maziar Zamanpour

✉ zaman050@umn.edu

☎ +1 4244683828

🏠 Maziar Zamanpour

🌐 maziar0z

EDUCATION

University of Minnesota - Minneapolis, MN Doctor of Philosophy (Ph.D.), Civil Engineering: Transportation (+ Industrial Engineering Minor)	Dec 2023 – Jan 2027 GPA 3.7
University of Minnesota - Minneapolis, MN Master of Science (M.S.), Civil Engineering: Transportation	Aug 2022 – Dec 2023 GPA 3.6
University of Tehran - Tehran Bachelor of Science (B.S.), Civil Engineering	Sep 2018 – May 2022 GPA 3.7

EXPERIENCE

Research Scientist — Analytical Modeling & Strategy for Connected Vehicle Systems

University of Minnesota - Minneapolis, MN	August 2022 - present
- Served as a key contributor on a 2-year engagement with the Minnesota Department of Transportation (MnDOT) as the client: scoping problems, designing solutions, managing deliverables, and presenting findings directly to agency decision-makers. - Coordinated across faculty, engineers, and MnDOT stakeholders to align project timelines, synthesize inputs from multiple perspectives, and deliver recommendations that balanced technical rigor with client needs. - Communicated complex analytical findings to diverse audiences through 7 publications, 2 client-facing technical reports, and presentations at major conferences (TRB, INFORMS, ITS Minnesota, MAASTO, Smart Cities). - Led the design, field testing, and client delivery of a real-time safety warning system for MnDOT: managed data collection on public roads in Scott County, MN, authored the final technical report, and presented results that directly informed the client's deployment decision (Demo). - Structured and solved a resource-constrained network design problem for pedestrian infrastructure, translating a vague city-planning challenge into a rigorous optimization framework that reduced computation time by 90% (GitHub). - Conducted cost-benefit analysis for MnDOT on connected vehicle corridor deployment; built the financial case showing payback within 3 years, directly shaping the client's infrastructure investment strategy. - Developed an energy-efficient vehicle control strategy achieving up to 13% cost savings, demonstrating ability to quantify impact and translate analytical results into actionable recommendations.	

Entrepreneurial Lead — Technology Strategy & Market Validation

National Science Foundation, National I-Corps Program - United States	May 2024 - Present
- Led a cross-functional team of researchers and business mentors to develop, test, and refine a technology product from concept through market validation. - Conducted 120+ structured customer discovery interviews and synthesized findings into a strategic product roadmap, iterating on the value proposition based on evidence from diverse user segments. - Developed an ML-based demand prediction and dispatch optimization system for a car-sharing platform, achieving modeled 75% cost savings—translating complex analytics into a clear business case. - Secured a \$50K grant by presenting a compelling case for the technology's market potential to a review panel. - Collaborated in a multi-industry cohort (finance, engineering, tech), sharing weekly progress presentations and exchanging cross-sector perspectives on innovation and client pain points.	

Independent Auto Sales (Part-Time Entrepreneurship)

Minneapolis, MN	August 2025 - present
- Built a small vehicle resale business from scratch with personal capital—handling sourcing, market analysis, repairs, pricing strategy, marketing, and negotiation through to close. - Developed strong client insight and relationship skills; repeat and referral customers demonstrate trust-based selling and ability to match solutions to client needs.	

Course Instructor — Probability and Statistics

University of Minnesota - Minneapolis, MN

August 2023 - May 2025

- Led weekly discussion sessions for 30+ students, designing materials that translated abstract statistical concepts into accessible, structured narratives.
- Covered probability, causal inference, Bayesian methods, regression, and estimation—building experience communicating quantitative topics to non-specialist audiences.

PROJECTS

Mobilislice: Personalized Car-Sharing Platform

Winter 2024 – present

- Designed and built a car-sharing product using ML-based trip prediction and optimization-driven vehicle dispatch (Python, CPLEX, SAP), balancing technical complexity with user-centric design.

DoorDash Delivery Assignment Optimization

Fall 2025

- Formulated a delivery-driver matching problem and designed both an optimal solution (integer programming) and a practical real-time heuristic, demonstrating ability to balance rigor with implementation constraints.
- Designed and executed SQL queries for data extraction, filtering, and customization to support analysis and inform the optimization model inputs.

PUBLICATIONS & CONFERENCES

- Traffic Prediction-Based Individualized Driver Warning System to Reduce Red Light Violations, *Journal of Transportation Engineering, Part A: Systems*, 2026.

- Incorporating lane-change prediction into energy-efficient speed control of connected autonomous vehicles at intersections, *Transportation Research Part C: Emerging Technologies*, Volume 171, 2025 (also presented at 2023 INFORMS).

- Development and Demonstration of a Novel Red Light Running Warning System Using Connected V2I Technology, *Minnesota Department of Transportation*, 2024.

- Cost/Benefit Analysis of Fuel-Efficient Speed Control Using SPaT Data, *Minnesota Department of Transportation*, 2023.

- Three additional papers submitted/under review; presented at 2024 INFORMS (Seattle, WA) and 2024 Transportation Research Board.

SKILLS

- **Problem Solving & Analytics:** Structured problem decomposition, data-driven decision-making, cost-benefit analysis, optimization modeling (Python, Gurobi, CPLEX), statistical analysis, predictive modeling.
- **Communication:** Storytelling with data, stakeholder presentations, technical report writing, translating complex findings for non-technical audiences.
- **Programming & Tools:** Python (Pandas, NumPy, scikit-learn), Java, R, SQL, Tableau, SAP, Excel, PowerPoint, Google Workspace.
- **Leadership:** Cross-functional team coordination, project management, mentorship, customer discovery, multi-stakeholder alignment.

AWARDS & RECOGNITION

- Won Matthew J. Huber Award for excellence in transportation research and development (2024).
- Won Transportation Research Board Travel Award two times (2023 and 2025).
- Entrusted with 4 years of continuous project funding from the Minnesota Department of Transportation and National Science Foundation.
- Recognized among the top graduate students in the Civil Engineering department's annual magazine (2024).
- ITS (Intelligent Transportation Systems) Minnesota member and presenter since 2024.

RELEVANT COURSES

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| • Storytelling for Engineers | • Analytics and Data-Driven Decision-Making |
| • Decision Analysis | • Operational Research for Data Science |
| • Production and Inventory Systems | • Optimization for Machine Learning |